INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHAGALPUR
MA203 (Probability and Statistics)
MIDSEM 2022-23 Spring

Max. Marks: 30 **Time:** 2.00 hours **Date of Exam:** 10-03-2023

1)

a) Prove that $P(\bar{A}|B) = 1 - P(A|B)$ *[3 marks]*

b) A box Contains 6 red, 4 white and 5 black balls. A person draws 4 balls from the box, at random. Find the probability that among the balls drawn there is at least one ball of each color. *[4 marks]*

c) While answering a multiple choice question (MCQ), a student either knows the answer with probability p or he/she makes a guess with probability $1-p$. If four choices are given in an MCQ, the probability that the student got the correct answer while guessing is $\frac{1}{4}$. What is the probability that a student actually knows the answer to an MCQ given he/she got the answer correct? *[3 marks]*

2)

a) Find the probability mass function $f(x)$ of a random variable X, where the cumulative distribution function $F(x)$ is given by *[2 marks]*

$$F(x) = \begin{cases} 0 & -\infty < x < 0 \\ 1/5 & 0 \le x < 1 \\ 3/5 & 1 \le x < 2 \\ 1 & 2 \le x < \infty \end{cases}$$

b) The joint density function of two discrete random variables X and Y are given as:

$$f(x,y) = \frac{\lambda^x e^{-\lambda} p^y (1-p)^{x-y}}{y!(x-y)!}$$ *[3 marks]*

Show that the marginal density function $f_X(x) = \frac{\lambda^x e^{-\lambda}}{x!}$

c) Show that $Y = \frac{(X-\mu)^2}{\sigma^2}$ has a chi-squared distribution with 1 degree of freedom, when X has a normal distribution with mean μ and variance σ^2.

[5 marks]

3)

a) The probability that a product is defective from a certain factory $\frac{2}{5}$. If 110 products are produced from that factory, what is the probability that fewer than 70 products are good? *(Hint: normal approx. of binomial)* *[3 marks]*

b) Find the moment generating function of the uniform distribution on the interval [0,1]. *[2 marks]*

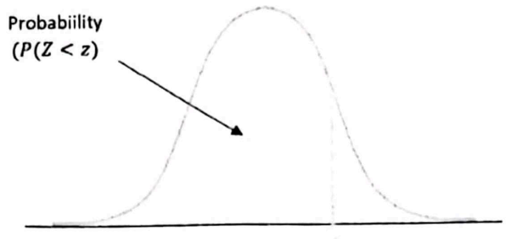
c) Prove that for any continuous random variable X, the cumulative distribution function $F(x)$ satisfies $F(a) \geq \frac{a-\mu_x}{a}$, where μ_x the mean of the random variable is X.
[2 marks]

d) Traveling between two campuses of a university in a city via shuttle bus takes, on average, 28 minutes with a standard deviation of 5 minutes. In a given week, a bus transported passengers 40 times. What is the probability that the average transport time was more than 30 minutes? Assume the mean time is measured to the nearest minute. *[3 marks]*

---------------------X---------------------(standard normal table next page)

$Y - \frac{x-\mu}{\sigma}$, $\frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$ 28 P

$29.5 < \bar{x} > 30.5$



	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5754
	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
	0.7258	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
	0.7580	0.7612	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
	0.7881	0.7910	0.7939	0.7967	0.7996	0.8023	0.8051	0.8079	0.8106	0.8133
	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9430	0.9441
	0.9452	0.9463	0.9474	0.9485	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9700	0.9706
	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9762	0.9767
	0.9773	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
	0.9861	0.9865	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9980	0.9980	0.9981
	0.9981	0.9982	0.9983	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998	0.9998